

Exploratory Data Analysis

using scikit-learn

INTRO



built-in machine algorithms and models

why

- comes with classic sample data sets
- Simple tools for predictive data analysis
- Accessible and reusable in various contexts
- Built on NumPy, SciPy, and matplotlib

packages

```
python -m venv venv  
./venv/bin/pip install scikit-learn
```

alternative installation using conda, active state or
system packages

starting

```
import sklearn  
print(sklearn.__version__)
```

1.6.1

IRIS DATASET



loading

```
import sklearn  
  
iris = sklearn.datasets.load_iris()  
  
# print(iris)  
print(iris.keys())
```

```
dict_keys(['data', 'target', 'frame', 'target_names', 'DESCR', 'feature'
```

DESCR

```
import sklearn.datasets
```

```
iris = sklearn.datasets.load_iris()
```

```
print(iris.DESCR)
```

```
.. _iris_dataset:
```

```
Iris plants dataset
```

```
-----
```

```
**Data Set Characteristics:**
```

```
:Number of Instances: 150 (50 in each of three classes)
```

```
:Number of Attributes: 4 numeric, predictive attributes and the class
```

```
:Attribute Information:
```

- sepal length in cm
- sepal width in cm
- petal length in cm
- petal width in cm
- class:
 - Iris-Setosa
 - Iris-Versicolour
 - Iris-Virginica

:Summary Statistics:

	Min	Max	Mean	SD	Class Correlation
sepal length:	4.3	7.9	5.84	0.83	0.7826
sepal width:	2.0	4.4	3.05	0.43	-0.4194
petal length:	1.0	6.9	3.76	1.76	0.9490 (high!)
petal width:	0.1	2.5	1.20	0.76	0.9565 (high!)

:Missing Attribute Values: None

:Class Distribution: 33.3% for each of 3 classes.

:Creator: R.A. Fisher

:Donor: Michael Marshall (MARSHALL%PLU@io.arc.nasa.gov)

:Date: July, 1988

The famous Iris database, first used by Sir R.A. Fisher. The dataset is from Fisher's paper. Note that it's the same as in R, but not as in the Machine Learning Repository, which has two wrong data points.

This is perhaps the best known database to be found in the pattern recognition literature. Fisher's paper is a classic in the field and is referenced frequently to this day. (See Duda & Hart, for example.) The data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant. One class is linearly separable from the other 2;

latter are NOT linearly separable from each other.

.. dropdown:: References

- Fisher, R.A. "The use of multiple measurements in taxonomic problems" *Annual Eugenics*, 7, Part II, 179-188 (1936); also in "Contributions to Mathematical Statistics" (John Wiley, NY, 1950).
- Duda, R.O., & Hart, P.E. (1973) *Pattern Classification and Scene Analysis* (Q327.D83) John Wiley & Sons. ISBN 0-471-22361-1. See page 218.
- Dasarathy, B.V. (1980) "Nosing Around the Neighborhood: A New System Structure and Classification Rule for Recognition in Partially Explored Environments". *IEEE Transactions on Pattern Analysis and Machine Intelligence*, Vol. PAMI-2, No. 1, 67-71.
- Gates, G.W. (1972) "The Reduced Nearest Neighbor Rule". *IEEE Transactions on Information Theory*, May 1972, 431-433.
- See also: 1988 MLC Proceedings, 54-64. Cheeseman et al's AUTOCLASS conceptual clustering system finds 3 classes in the data.
- Many, many more ...

classification

```
print(iris.target_names)
```

```
['setosa' 'versicolor' 'virginica']
```



data

```
import sklearn.datasets
```

```
iris = sklearn.datasets.load_iris()
```

```
print(type(iris.data))
```

```
print(iris.data.shape)
```

```
print(iris.feature_names)
```

```
print(iris.target_names)
```

```
print(iris.data[:5])
```

```
<class 'numpy.ndarray'>
```

```
(150, 4)
```

```
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']
```

```
['setosa' 'versicolor' 'virginica']
```

```
[[5.1 3.5 1.4 0.2]
```

```
 [4.9 3.  1.4 0.2]
```

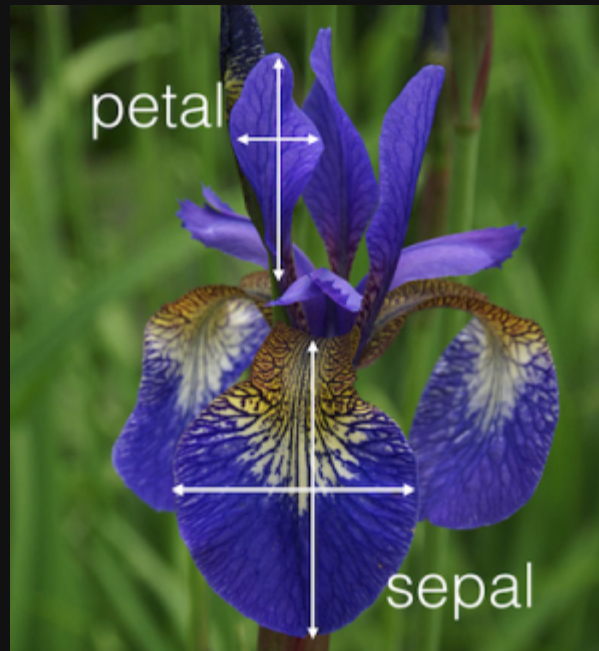
```
 [4.7 3.2 1.3 0.2]
```

```
 [4.6 3.1 1.5 0.2]
```

```
 [5.  3.6 1.4 0.2]]
```

measurements

by Edgar Anderson



150 measurement sample of petal and sepal

history

Sir Ronald Fisher, in his 1936 eugenics paper

The use of multiple measurements in taxonomic problems

used iris dataset as an example for **linear discriminant analysis**

to accurately distinguish 3 species only using sepal and petal measurements.

problem 1

mean of each measurements

```
[5.84333333 3.05733333 3.758      1.19933333]
```

problem 1

mean of each measurements

```
import sklearn
```

```
iris = sklearn.datasets.load_iris()
```

```
print(iris.data.mean(axis=0))
```

```
[5.84333333 3.05733333 3.758      1.19933333]
```


problem 2

std of each measurements

```
sepal length (cm) : 0.83  
sepal width (cm)  : 0.43  
petal length (cm) : 1.76  
petal width (cm)  : 0.76
```

problem 2

std of each measurements

```
import sklearn

iris = sklearn.datasets.load_iris()
mean = iris.data.std(axis=0)

for k, v in zip(iris.feature_names, mean):
    print('{:17} : {:.2f}'.format(k, v))
```

```
sepal length (cm) : 0.83
sepal width (cm)  : 0.43
petal length (cm) : 1.76
petal width (cm)  : 0.76
```

problem 3

show percentile

[0.2 6.7]

problem 3

show percentile

```
import sklearn
import numpy as np

iris = sklearn.datasets.load_iris()
print(np.percentile(iris.data, q=[5, 95]))
```

```
[0.2 6.7]
```

problem 4

what is the species having max sepal length

```
131  
[7.9 3.8 6.4 2. ]  
2  
virginica
```

```
for (i, (s1, *_)), target in zip(enumerate(iris.data), iris.target):  
    if s1 >= high:  
        high = s1  
        print(s1, i, target, iris.target_names[target])
```

problem 4

what is the species having max sepal length

```
import sklearn
```

```
iris = sklearn.datasets.load_iris()  
print(i:=iris.data[:, 0].argmax())  
print(iris.data[i])  
print(t:=iris.target[i])  
print(iris.target_names[t])
```

```
131
```

```
[7.9 3.8 6.4 2. ]
```

```
2
```

```
virginica
```

```
for (i, (sl, *_)), target in zip(enumerate(iris.data), iris.target):  
    if sl >= high:  
        high = sl  
        print(sl, i, target, iris.target_names[target])
```

problem 5

separate mean of each section

	setosa	versicolor	virginica
sepal length (cm) :	5.01	5.94	6.59
sepal width (cm) :	3.43	2.77	2.97
petal length (cm) :	1.46	4.26	5.55
petal width (cm) :	0.25	1.33	2.03

```
import sklearn

iris = sklearn.datasets.load_iris()
s1 = iris.data[iris.target == 0].mean(axis=0)
s2 = iris.data[iris.target == 1].mean(axis=0)
s3 = iris.data[iris.target == 2].mean(axis=0)
mean = iris.data.mean(axis=0)
print(' '*19, *iris.target_names)
for k, v1, v2, v3 in zip(iris.feature_names, s1, s2, s3):
    print('%-17s : %3.2f %6.2f %10.2f'%(k, v1, v2, v3))
```


IRIS DATAFRAME

lets use pandas

dataframe

```
import sklearn
import pandas as pd

iris = sklearn.datasets.load_iris()
df = pd.DataFrame(
    data = iris.data,
    columns = iris.feature_names,
)
print(df.head())
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width
0	5.1	3.5	1.4	
1	4.9	3.0	1.4	
2	4.7	3.2	1.3	
3	4.6	3.1	1.5	
4	5.0	3.6	1.4	

df.describe()

```
print(df.describe())
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.198667
std	0.828066	0.435866	1.765298	0.435322
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	0.400000
75%	6.400000	3.300000	5.100000	0.500000
max	7.900000	4.400000	6.900000	0.600000

columns names

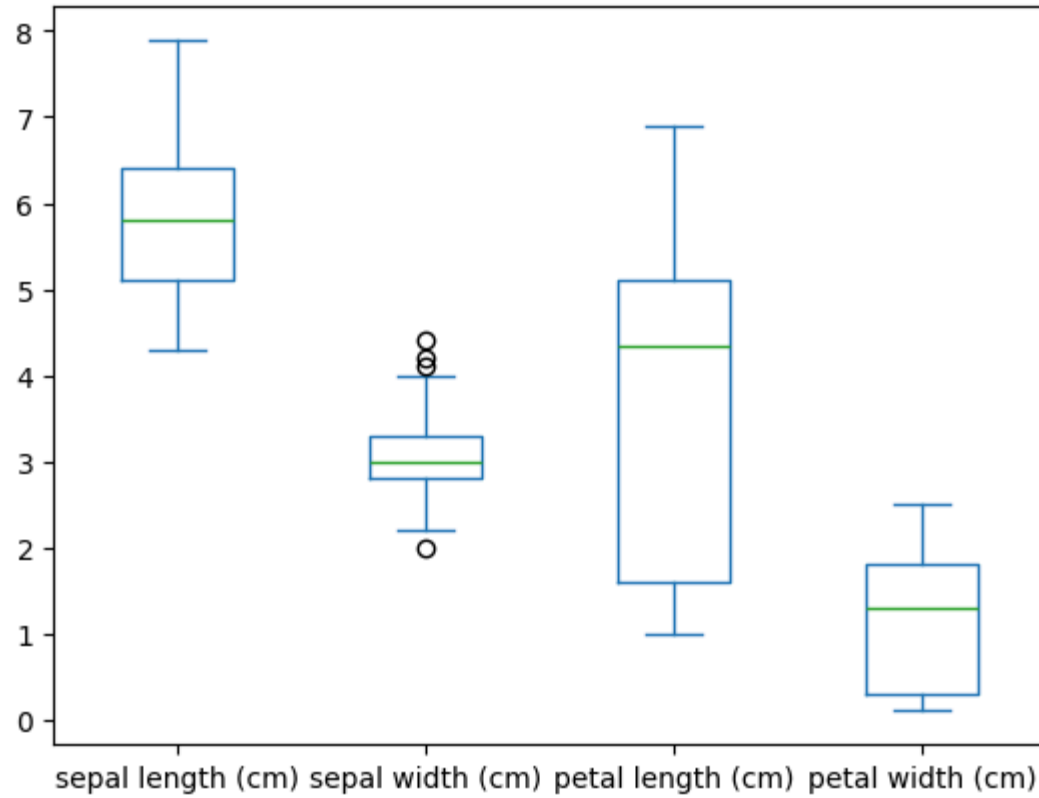
```
import sklearn
import pandas as pd

iris = sklearn.datasets.load_iris()
df = pd.DataFrame(
    data      = iris.data,
    columns   = iris.feature_names,
)
df.columns = [ 'sl', 'sw', 'pl', 'pw' ]
print(df.head())
```

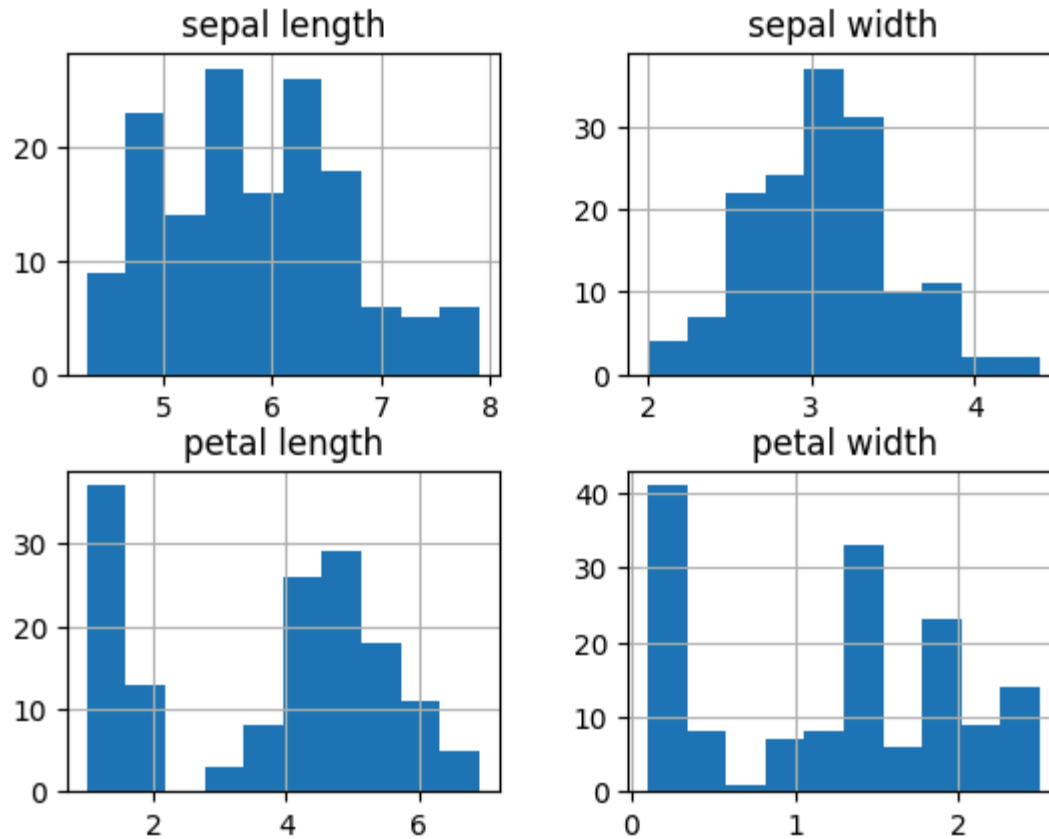
	sl	sw	pl	pw
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

df.plot.box()

```
import matplotlib.pyplot as plt  
  
ax = df.plot.box().get_figure()  
plt.show()
```



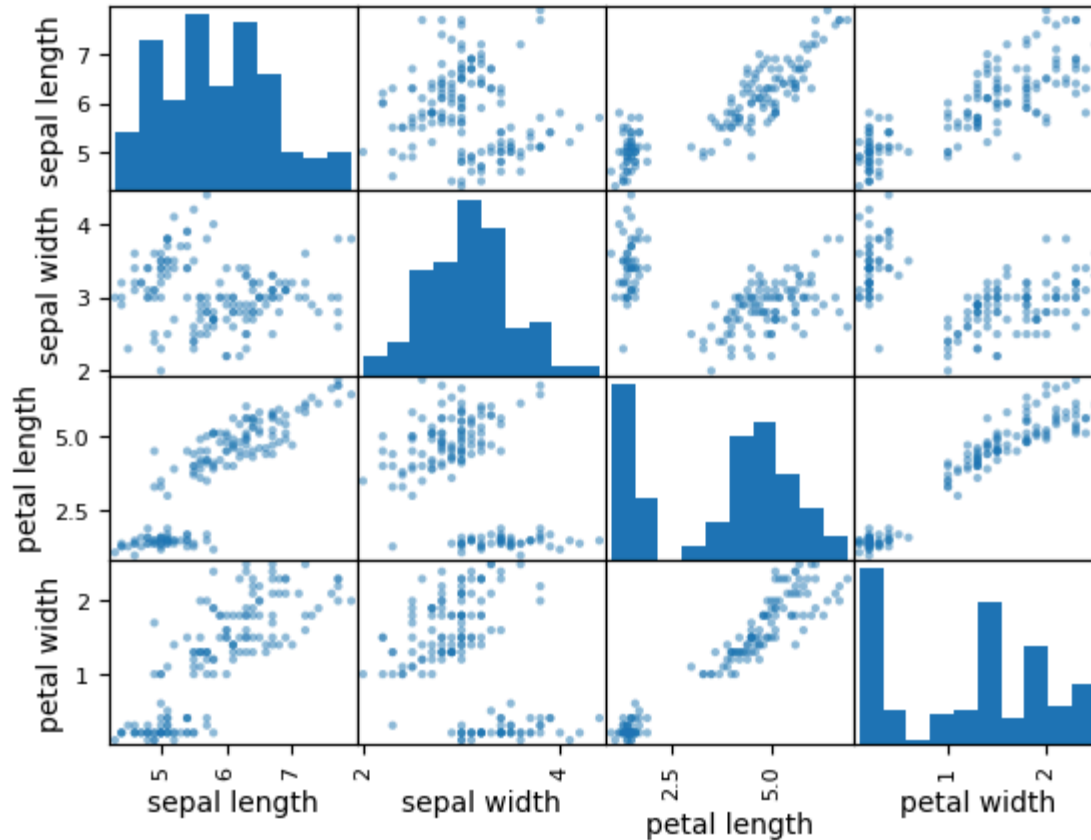
df.hist()



```
import matplotlib.pyplot as plt
import sklearn
import pandas as pd

iris = sklearn.datasets.load_iris()
df = pd.DataFrame(
    data      = iris.data,
    columns   = iris.feature_names,
)
ax = df.hist()
plt.show()
```


plotting.scatter_matrix()

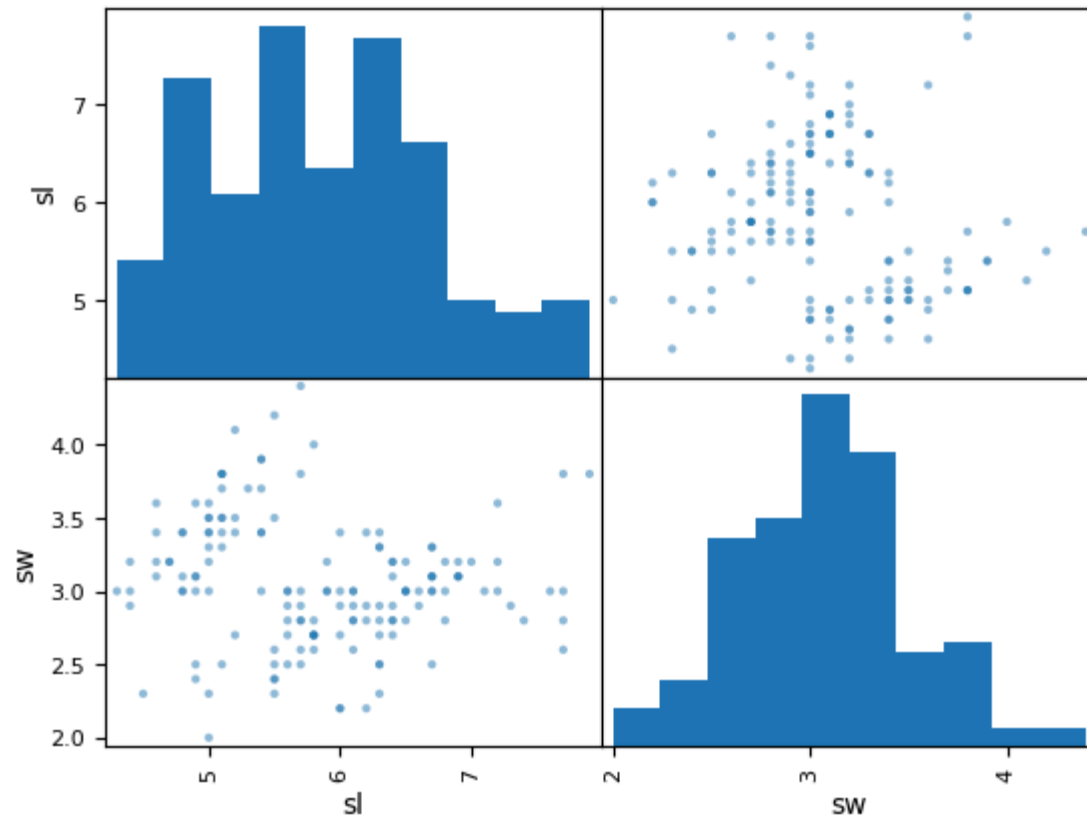


```
import matplotlib.pyplot as plt
import sklearn
import pandas as pd

iris = sklearn.datasets.load_iris()
df = pd.DataFrame(
    data      = iris.data,
    columns   = iris.feature_names,
)
ax = pd.plotting.scatter_matrix(df)
plt.show()
```

problem 1

plot the scatter matrix dataset



```
import matplotlib.pyplot as plt
import sklearn
import pandas as pd

iris = sklearn.datasets.load_iris()
df = pd.DataFrame(
    data      = iris.data,
    columns   = iris.feature_names,
)
df.columns = ['sl', 'sw', 'pl', 'pw']
ax = pd.plotting.scatter_matrix(df[['sl', 'sw']])
plt.show()
```

add target column

```
import sklearn
import pandas as pd

iris = sklearn.datasets.load_iris()
df = pd.DataFrame(
    data      = iris.data,
    columns   = iris.feature_names,
)
df['target'] = iris.target
print(df)
```

	sepal length	sepal width	petal length	petal width	target
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	2
146	6.3	2.5	5.0	1.9	2
147	6.5	3.0	5.2	2.0	2
148	6.2	3.4	5.4	2.3	2
149	5.9	3.0	5.1	1.8	2

```
[150 rows x 5 columns]
```

groupby

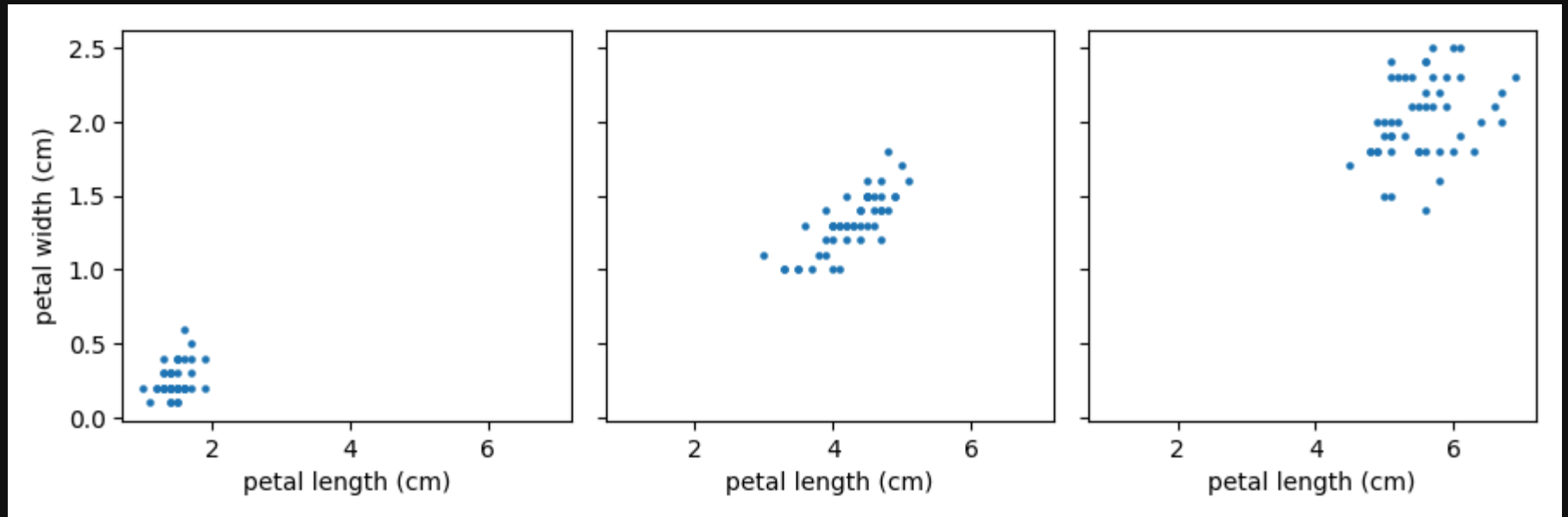
```
import sklearn
import pandas as pd

iris = sklearn.datasets.load_iris()
df = pd.DataFrame(
    data = iris.data,
    columns = iris.feature_names,
)
df['target'] = iris.target
print(df.groupby('target').mean())
```

	sepal length	sepal width	petal length	petal width
target				
0	5.006	3.428	1.462	0.246
1	5.936	2.770	4.260	1.326
2	6.588	2.974	5.552	2.026

problem 3

plot relation sepal length and width all species




```

import pandas as pd
import sklearn.datasets
import matplotlib.pyplot as plt

iris = sklearn.datasets.load_iris()
df = pd.DataFrame(
    data = iris.data,
    columns = iris.feature_names
)
df['species'] = list(map(lambda x: iris.target_names[x], iris.target))

fig, ax = plt.subplots(1, 3, layout='tight', sharex=True, sharey=True)
for i, (name, group) in enumerate(df.groupby('species')):
    group.plot.scatter(ax=ax[i], x='petal length (cm)', y='petal width (cm)')

```